RESEARCH COMPANION

Moral alignment benchmarks

Annotated bibliography for *The Trajectory of Outer Alignment in Large Language Models*

Thirteen primary and adjacent sources, critically annotated for contribution, relevance, evidence status, and limitations.

October 11, 2026

Prepared from the report's works cited

Read the evidence as a portfolio, not a leaderboard

No single benchmark establishes moral competence. The most defensible reading combines complementary tests of process, stability, framing, cultural context, training, and system behavior.

QUESTION	BEST ANCHORS	WHAT THE PAIR REVEALS
Can the model reason procedurally?	MoReBench + Zhu & Lazar	Capability and measurement sensitivity
Does reasoning survive context shifts?	Staircase + CLASH + OMIT	Path dependence, perspective, framing
Which values and norms appear?	MoralBench + NormBank	Profiles and contextual social expectations
Can training and scaffolds reduce risk?	ECT + AgentBreeder	Feedback quality and system architecture
What should alignment optimize?	Positive + Bidirectional alignment	Flourishing, agency, institutions, co-adaptation

INTERPRETATION RULE 1

A benchmark score is evidence about performance under a protocol. It is not direct evidence of stable values, consciousness, or justified moral agency.

INTERPRETATION RULE 2

Reference rubrics are instruments, not ground truth. Audit their coverage, generality, cultural assumptions, and judge dependence.

INTERPRETATION RULE 3

Prefer triangulation: combine verdicts with reasoning traces, paired frames, escalating scenarios, repeated trials, and out-of-distribution tests.

INTERPRETATION RULE 4

Separate descriptive norms from normative justification. Crowd agreement can improve coverage while still encoding parochial or unjust preferences.

Scope. Entries prioritize primary sources that materially support the report's argument. Mirrors, repositories, profiles, and duplicate records are handled in the citation audit rather than counted as independent evidence.

01

PROCEDURAL REASONING

MoReBench: Evaluating Procedural and Pluralistic Moral Reasoning in Language Models, More than Outcomes

Chiu, Y. Y., Lee, M. S., Calcott, R., Handoko, B., de Font-Reaulx, P., Millière, R., et al. (2025; revised 2026). arXiv:2510.16380. **Access source**

PREPRINT; REVISED JUNE 2026

Contribution. Introduces 1,000 moral scenarios with more than 23,000 expert-authored criteria, plus a 150-case theory subset spanning five normative frameworks. It evaluates identified considerations, trade-off handling, recommendations, and harmfulness rather than only final verdicts.

Why it matters here. This is the report's central source for the move from outcome scoring to procedural and pluralistic evaluation. Its framework makes disagreements inspectable and supports analysis of model partiality toward specific ethical traditions.

Critical use note. Expert rubrics improve transparency but do not eliminate normative choice. Scores still depend on elicitation, rubric coverage, and automated judging; results should be paired with robustness checks and alternative rubrics.

02

PROCEDURAL REASONING

Are LLMs Bad at Moral Reasoning?

Zhu, M., & Lazar, S. (2026). arXiv:2606.11635. **Access source**

PREPRINT, JUNE 2026

Contribution. Reuses MoReBench in an "apples-to-apples" rubric-writing task. The authors report that frontier models recover 83–89% of human-listed considerations, add at least 2.26 times as many unique relevant considerations, and gain 18.4 points when non-general human criteria are rewritten.

Why it matters here. This is the essential counterweight to pessimistic readings of MoReBench. It shows that benchmark scores can conflate latent competence with response format, recognition rules, and defects in the reference rubric.

Critical use note. The paper is a reanalysis, not an independent benchmark, and its conclusions depend on model-generated rubrics and judge-mediated matching. Treat it as evidence about measurement sensitivity, not proof of general moral competence.

03

DYNAMIC ROBUSTNESS

The Staircase of Ethics: Probing LLM Value Priorities through Multi-Step Induction to Complex Moral Dilemmas

Wu, Y., Sheng, Q., Wang, D., Yang, G., Sun, Y., Wang, Z., Bu, Y., & Cao, J. (2025). Proceedings of EMNLP 2025, 15950–15970. **Access source**

PEER-REVIEWED CONFERENCE PAPER

Contribution. Presents 3,302 five-stage dilemmas and evaluates nine models as stakes and value conflicts escalate. The design exposes path dependence, preference reversals, and non-transitivity that one-shot moral questions cannot reveal.

Why it matters here. The paper substantiates the report's claim that ethical stability must be tested longitudinally. It is especially useful for stress-testing whether a model preserves or explicitly revises principles as context accumulates.

Critical use note. Scenario generation and value mapping involve automated components, so observed reversals may combine genuine instability with artifacts of the prompts, labels, or escalation template. Replication with human-authored trajectories would strengthen the inference.

04

DYNAMIC ROBUSTNESS

CLASH: Evaluating Language Models on Judging High-Stakes Dilemmas from Multiple Perspectives

Lee, A., Kwon, R. S., Railton, P., & Wang, L. (2025). arXiv:2504.10823v2. **Access source**

PREPRINT

Contribution. Curates 345 high-stakes dilemmas and 3,795 character perspectives. Across ten models, even leading systems score below 50% on ambivalent cases and degrade sharply when character values shift.

Why it matters here. CLASH operationalizes pluralism as perspective-sensitive judgment rather than a single consensus label. It supports the report's emphasis on ambivalence, stakeholder conflict, and temporal changes in values.

Critical use note. The benchmark tests fidelity to authored perspectives, not which perspective is ethically best. Domain selection, label construction, and narrative length can influence performance and limit generalization to live decisions.

05

FRAMING AND BIAS

OMIT the Action: Measuring Framing-Invariant Omission Bias under Philosophical Disagreement

Lee, S., Song, J., Kim, C., Kum, J., & Park, C. (2026). arXiv:2610.07847v1. **Access source**

VERY RECENT PREPRINT, OCTOBER 2026

Contribution. Builds 218 paired action/omission scenarios across ten conflict types using five normative perspectives. Eight models show omission bias, while prompts that elicit principles before a binary answer reduce that bias but may shift errors toward action bias.

Why it matters here. The mirror-frame design cleanly distinguishes a preference for inaction from a substantive preference over outcomes. It directly supports the report's warning that legacy safety tuning can produce passivity-like behavior.

Critical use note. This is a newly released preprint. Scenario transformation and persona-panel construction are model-assisted, and mitigation should be judged across the full response distribution rather than a single bias rate.

06

FRAMING AND BIAS

Conscience Conflict? Evaluating Language Models' Moral Understanding

Hota, A., & Jokinen, J. P. P. (2025). MoMLeT 2025, CEUR Workshop Proceedings, Vol. 4004. **Access source**

WORKSHOP PAPER

Contribution. Uses six human-authored, high-stakes vignettes requiring a choice and free-text justification. Five open-source models are coded for deontological, utilitarian, virtue, care, justice, pragmatic, and conflict-oriented rationales.

Why it matters here. The qualitative coding illustrates how acceptable decisions can conceal unstable or shallow rationales. It complements large benchmarks by making ethical style and coherence visible at case level.

Critical use note. The sample is intentionally small, so it cannot support broad model rankings or precise prevalence estimates. Coding judgments and prompt sensitivity should be treated as exploratory evidence.

07

MORAL PROFILES

MoralBench: Moral Evaluation of LLMs

Ji, J., Chen, Y., Jin, M., Xu, W., Hua, W., & Zhang, Y. (2024). arXiv:2406.04428v2. **Access source**

PREPRINT; OPEN BENCHMARK AND CODE

Contribution. Adapts Moral Foundations Theory to compare model responses across care, fairness, loyalty, authority, sanctity, and liberty. The benchmark emphasizes profile stability, comparative differences, and test–retest behavior.

Why it matters here. MoralBench provides an interpretable psychometric lens for auditing model-specific value patterns. It is useful as a forensic profile of post-training and data effects, not merely as a leaderboard.

Critical use note. Moral Foundations Theory is one contested account of morality. A stable profile does not establish justified reasoning, and the language of "moral identity" should not be mistaken for evidence that a model possesses agency or character.

08

MORAL PROFILES

NormBank: A Knowledge Bank of Situational Social Norms

Ziems, C., Dwivedi-Yu, J., Wang, Y.-C., Halevy, A., & Yang, D. (2023). Proceedings of ACL 2023, 7756–7776. **Access source**

PEER-REVIEWED CONFERENCE PAPER

Contribution. Provides 155,000 situational norms and 63,000 contextual constraints covering setting, roles, attributes, and sociocultural conditions. The resource models how small contextual changes can cancel or reverse an otherwise plausible norm.

Why it matters here. NormBank is the report's strongest source for bottom-up, context-rich descriptive alignment. It demonstrates the scale and conditionality needed to represent everyday social expectations.

Critical use note. The resource records what people judge to be normal or acceptable; it does not establish what is morally right. Sampling and cultural coverage can reproduce majority bias, and descriptive norms require normative review before use in high-stakes systems.

09

GOALS AND TRAINING

Positive Alignment: Artificial Intelligence for Human Flourishing

Laukkonen, R., Krier, S., Bakalar, C., Chandaria, S., Kringelbach, M., Elwood, A., et al. (2026). arXiv:2605.10310v3. **Access source**

POSITION PAPER / PREPRINT

Contribution. Defines positive alignment as building safe, cooperative systems that actively support human and ecological flourishing in pluralistic, context-sensitive, and user-authored ways. It proposes interventions across data, training, evaluation, agents, and governance.

Why it matters here. This paper supplies the report's normative shift from "not harmful" to constructively beneficial. It is valuable for setting evaluation targets that include autonomy, truth-seeking, resilience, and flourishing.

Critical use note. The agenda is conceptual rather than a validated benchmark. Operationalizing flourishing risks paternalism and metric gaming; user authorship, polycentric oversight, and explicit trade-off governance are necessary safeguards.

10

GOALS AND TRAINING

Evaluation-Conditioned Training: Teaching Models to Generalize to Stronger Oversight Regimes

Harris, A., Corra, K., Chaudhury, A., & Hao, Y. (2026). arXiv:2608.10209v1. **Access source**

PREPRINT; ACCEPTED TO A COLM 2026 WORKSHOP

Contribution. Conditions post-training samples on feedback fidelity, then requests behavior associated with a high-fidelity evaluator at deployment. Two proof-of-concept studies target political even-handedness and arithmetic sycophancy under deliberately imperfect feedback.

Why it matters here. ECT is relevant to the report's reward-misspecification argument because it attempts to separate desired behavior from the limitations of the evaluator that produced training signals.

Critical use note. The evidence is narrow and does not yet cover pluralistic moral dilemmas, long-horizon agents, or adversarial monitor labels. It should be viewed as a promising training primitive, not a solved route around weak oversight.

11

AGENTIC SYSTEMS

AgentBreeder: Mitigating the AI Safety Risks of Multi-Agent Scaffolds via Self-Improvement

Rosser, J., & Foerster, J. (2025). arXiv:2502.00757v4. **Access source**

CONFERENCE WORK; REVISED OCTOBER 2025

Contribution. Applies multi-objective evolutionary search to multi-agent scaffolds. “Blue” optimization improves safety-benchmark performance while preserving capability, whereas “red” optimization finds high-performing but adversarially weak scaffolds.

Why it matters here. The work supports the report’s key systems claim: alignment is a property of the model-plus-scaffold, not only model weights. It offers a practical red-team/blue-team framework for agent architecture selection.

Critical use note. Reported uplift is relative to selected benchmarks and scaffold seeds, not a guarantee of deployment safety. Evolutionary search may discover benchmark-specific behavior, and emergent interaction risks require independent, out-of-distribution evaluation.

12

AGENTIC SYSTEMS

Position: Towards Bidirectional Human-AI Alignment

Shen, H., Knearem, T., Ghosh, R., Alkiek, K., Krishna, K., Liu, Y., et al. (2025). NeurIPS 2025 Position Paper Track. **Access source**

NEURIPS 2025 POSITION PAPER

Contribution. Synthesizes more than 400 papers and frames alignment as a reciprocal process: integrating human values into AI while helping people critically understand, collaborate with, and adapt to AI. It identifies research gaps in long-term interaction, value modeling, and mutual understanding.

Why it matters here. This source broadens the report’s “full-stack” thesis beyond technical controls to human agency, institutions, literacy, and social feedback loops.

Critical use note. As a position paper, it organizes a field rather than demonstrating one intervention. “Aligning humans to AI” can be misread as compliance; the useful interpretation is capability-building and calibrated oversight that preserves human choice.

13

ADJACENT INNER ALIGNMENT

Goal Multi-Stage Verification: A Defense-in-Depth Mathematical Framework for Mitigating Deceptive Alignment in Large Language Models

Anonymous authors. (2026). Under review at MathAI 2026. **Access source**

ANONYMOUS SUBMISSION UNDER REVIEW

Contribution. Proposes a composite pipeline combining process-reward evaluation, formal verification, and multiple safety gates to increase the difficulty of passing with disconnected or strategically misleading reasoning.

Why it matters here. The paper is relevant background for the report's distinction between outer and inner alignment and for its call to test reasoning across several representational layers.

> **Critical use note.** It is not a normative-ethics benchmark, is not yet peer reviewed, and uses mathematical-reasoning accuracy as an indirect proxy for deceptive-alignment mitigation. Do not treat those gains as evidence that hidden objectives have been detected or corrected.

What survives cross-source comparison

- **Process is necessary but not sufficient.** Procedural rubrics reveal missing considerations, yet the rubric and elicitation method can dominate the score.
- **Stability requires dynamic tests.** Escalation, value shifts, and mirror framing uncover failures hidden by static prompts.
- **Pluralism must be explicit.** Ethical frameworks, stakeholder perspectives, and cultural context should be represented rather than averaged into one label.
- **Evaluation must extend to the full system.** Multi-agent scaffolds, monitoring, interfaces, and human adaptation can create or mitigate alignment failures.

The report's bibliography needs consolidation before reuse

The underlying report identifies the right research clusters, but its works-cited list mixes primary papers, mirrors, repositories, profiles, and at least one weakly related teaching page. The cleaned list below avoids double-counting evidence.

USE AS PRIMARY EVIDENCE

Peer-reviewed venue pages or direct paper records: MoReBench; Staircase of Ethics; CLASH; OMIT; Conscience Conflict; MoralBench; NormBank; Positive Alignment; ECT; AgentBreeder; Bidirectional Human-AI Alignment.

USE WITH HEIGHTENED CAUTION

Are LLMs Bad at Moral Reasoning? is a valuable preprint reanalysis, not an independent benchmark. The anonymous MathAI submission is under review and addresses inner-alignment verification only indirectly.

DO NOT COUNT TWICE

Scale Labs, alphaXiv, Hugging Face Papers, the MoReBench dataset page, and the NormBank GitHub repository are mirrors, explainers, or artifacts supporting the same underlying works.

OMIT FROM EVIDENTIARY CLAIMS

Author profile pages are discovery tools, not sources. The cited Stanford CS186 representation-learning lecture does not substantiate claims about moral benchmarks and should not be used for that purpose.

Recommended evaluation stack

1. **Coverage**
   Expert and model-generated rubrics, compared explicitly.
2. **Robustness**
   Paraphrases, mirror frames, repeated trials, and escalating stakes.
3. **Pluralism**
   Multiple ethical theories, cultures, languages, and stakeholders.
4. **System safety**
   Model, scaffold, monitoring, interface, and governance as one unit.

BOTTOM LINE

The report's strongest conclusion is methodological: evaluate how models reason, how stable that reasoning remains, whose values are represented, and what the surrounding system does. Treat every benchmark as one lens with known blind spots.